1/10/2026

# Practical 14

Write a code using RAW sockets to implement packet sniffing

Aim:

To write a code using RAW sockets to implement packet sniffing

Program:

```
from scapy.all import sniff
from scapy.layers.inet import IP
def packet_callback(packet):
    if IP in packet:
        ip_layer = packet[IP]
        protocol = ip_layer.proto
        src_ip = ip_layer.src
        dst_ip = ip_layer.dst
        protocol_name = ""
        if protocol == 1:
            protocol_name = "ICMP"
        elif protocol == 6:
            protocol_name = "TCP"
        elif protocol == 17:
            protocol_name = "UDP"
        else:
            protocol_name = "Unknown Protocol"
        print(f"Protocol: {protocol_name}")
        print(f"Source IP: {src_ip}")
        print(f"Destination IP: {dst_ip}")
        print("=" * 50)

sniff(
    iface = 'MediaTek Wi-Fi 6E MT7902 Wireless LAN Card
    prn = packet_callback,
    filter = "ip",
    store = 0,
    count = 10
)
```

Output:

```
C:\Users\Vedhasree S\Desktop\12 13 14 Experiments>python packet_sniffing.py
Protocol: UDP
Source IP: 192.168.29.115
Destination IP: 239.255.255.250
==================================================
Protocol: TCP
Source IP: 192.168.29.115
Destination IP: 40.79.150.127
==================================================
Protocol: TCP
Source IP: 192.168.29.115
Destination IP: 40.79.150.127
==================================================
Protocol: TCP
Source IP: 192.168.29.115
Destination IP: 40.79.150.127
==================================================
Protocol: TCP
Source IP: 192.168.29.115
Destination IP: 40.79.150.127
==================================================
Protocol: TCP
Source IP: 192.168.29.115
Destination IP: 4.150.223.100
==================================================
Protocol: TCP
Source IP: 40.79.150.127
Destination IP: 192.168.29.115
==================================================
Protocol: TCP
Source IP: 40.79.150.127
Destination IP: 192.168.29.115
==================================================
Protocol: TCP
Source IP: 40.79.150.127
Destination IP: 192.168.29.115
==================================================
Protocol: TCP
Source IP: 192.168.29.115
Destination IP: 40.79.150.127
==================================================
```

Result:

Thus, packet sniffing was successfully implemented using RAW sockets/Scapy to capture and display network packets with their protocol, source IP and destination IP.